The medial temporal lobe and the retention of scene memories across a one-week delay

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ABSTRACT

Remembering everyday events involves retaining multiple components, including the scenes that provide the spatial and contextual setting in which events occur. The medial temporal lobe (MTL) supports scene construction and complex event memory. However, less is known about its contribution to retaining the details of an event's scene over time, as scene information is typically embedded within narrative and event structures that are assessed for memory shortly after encoding. To examine scene memory retention, fifty-seven cognitively healthy older adults viewed 360-degree panoramic videos depicting real-world environments and freely recalled each scene shortly after viewing and again one week later. We tested whether individual differences in MTL volume were related to retention of scene memory and whether these relationships changed across the one-week delay. The proportion of episodic detail retrieved did not significantly change across the one-week delay, but the details recalled changed and became more participant specific, as measured by several measures of text-based similarity. Critically, relationships between MTL volume and scene memory also changed over time such that greater MTL volume was associated with being more consistent with what was originally recalled. Exploratory subregional analyses further showed that perirhinal cortex volume was robustly associated with the text-based similarity measures. These findings suggest that MTL is related to memory for the scene component of complex experience, particularly the ability to consistently recall the details of a scene over time.

Keywords: *Medial temporal lobe, scene memory, volume, episodic memory*

INTRODUCTION

Remembering everyday events often involves retrieving a complex combination of details about the people, objects, places, and activities that comprise them. Traditional laboratory paradigms, however, typically reduce event memory stimuli to words, images, or objects in ways that may limit what they can reveal about memory for complex everyday experiences. Researchers have therefore increasingly turned to using naturalistic approaches that capture the richness of daily events (Rubin & Umanath, 2015; Zacks, 2020), including assessing memory for story narratives (Lee et al., 2020; Cohn-Sheehy et al., 2022; Delarazan et al., 2026a) and films that contain complex visuals and dialogue (Delarazan et al., 2023; Fenerci et al., 2024; Delarazan et al., 2026b), as a means to complement the foundational insights revealed by traditional laboratory experiments. The present study builds on this naturalistic approach to event memory research, with a particular focus on understanding how the medial temporal lobe (MTL) may support memory for the scenes in which complex events occur.

Scenes are an essential component of complex everyday events, serving as a spatial and contextual stage on which people, objects, and activities can be perceived and reconstructed in a coherent memory (Bar, 2004; Hassabis & Maguire, 2007; Robin et al., 2018; Robin, 2019; Rubin & Umanath, 2015). A substantial body of neuropsychological and neuroimaging research implicates the MTL, particularly the hippocampus and parahippocampal cortex, in scene memory, supporting not only perception (Byrne et al., 2007; Bird and Burgess, 2008) and mental scene construction (Hassabis et al., 2006; Irish et al., 2017), but also the retrieval of scene-related details and maps (O'Keefe and Nadel, 1978; Cabeza et al., 2004). While these studies highlight a connection between the MTL and scene memory, questions remain about the nature of this brain-behavior relationship.

Specifically, an unresolved question is how the MTL supports the retention of details about complex scenes. Much of the existing literature has focused on perception and recognition paradigms that do not directly assay memory for the details of the scene (Eichenbaum et al., 2007; Lee et al., 2005; Staresina et al., 2012), or mental scene construction paradigms that do not test memory for recently encoded information (Hassabis et al., 2007; Schacter et al., 2012; Sheldon & Levine, 2016). Studies that have examined the details for recently encoded information have used films and narratives that contain sequences of events, characters, dialogue, and developing storylines (Lee & Chen, 2020; Antony et al., 2024; Delarazan et al., 2026a), which makes it difficult to determine how scene-specific information is retained independently of other components of an event, such as event order and narrative structure. Prior research using naturalistic approaches has indicated that the MTL may be an important neural network hub for multiple features of event memory. For instance, studies using films, narratives, and autobiographical memory suggest that the MTL may detect event boundaries (Reagh et al., 2020; Bailey et al., 2013), encode and retrieve a range of episodic and semantic details (Cabeza et al., 2004; Staresina et al., 2013), and combine information into a unified memory representation (Reagh & Ranganath, 2023). Studying the recall of details from recently encoded naturalistic scenes, in the absence of narratives and unfolding events, could therefore provide clarity on the role of the MTL in detailed scene retention.

Understanding the role of the MTL in scene memory for complex events also requires considering how this relationship may change over time, as the content of a memory evolves. It is well established that what is available for retrieval after several days often differs from what can be accessed immediately after an experience (Sekeres et al., 2016; Oedekoven et al., 2017). Prior research also suggests that, over time, memories do not simply become uniformly weaker

(Frankland & Bontempi, 2005; Winocur & Moscovitch, 2011; Squire et al., 2015). According to trace transformation theory, the MTL has a critical role in retaining central details of event memories, even if these details, and the role of the MTL, evolve over time (Wincour & Moscovitch, 2011). Notably, a prior study using short stories found that MTL volume was associated with how much detail for the stories was retained after several days of delay (Saloner et al., 2018). Interestingly, there was no relationship between MTL volume and immediate story recall, suggesting that, over time, the importance of the MTL to the retention of detailed event narratives may become more apparent. Whether similar outcomes apply to the retention of naturalistic scene details remains unclear.

In the present study, we sought to examine how individual differences in MTL volume relates to the retention of scene details across a one-week delay. We developed a novel behavioral task where participants viewed a series of 360-degree panoramic videos depicting real-world environments and freely recalled each scene shortly after viewing and again after a one-week delay. We selected 360-degree panoramic videos because they capture naturalistic scenes without the narrative structure or unfolding sequence of events present in films and stories. We tested whether among a group of older adults, individual differences in MTL volume were associated with individual differences in scene memory, and whether these associations changed over time. Critically, we characterized retention in complementary ways. Detail-based scoring captured the episodic composition of recall, whereas text-based similarity measures captured the consistency of remembered scene content relative to the original scene, what others recalled, and an individual's own recall across two sessions. Together, these measures allowed us to distinguish the overall composition of recall from specific scene information that remained represented across the delay. Finally, because different MTL subregions may make distinct contributions to complex

event memory, we conducted exploratory analyses examining relationships with hippocampal, parahippocampal, perirhinal, and entorhinal cortex volume.

Consistent with the link between the MTL and complex event memory (Rubin, 2006; Bailey et al., 2013), along with the theorized importance of this region to scene perception and memory (Hassabis & Maguire, 2007; Lee et al., 2005), we hypothesized that variability in MTL volume would be associated with the retention of detailed scene memories. Given evidence that both the content of memory and its relationship with MTL can change across retention intervals (Saloner et al., 2018), we further expected that the MTL-scene memory association would evolve over the course of the one-week delay, with the MTL maintaining a relationship with the retention of scene memory at delayed retrieval.

METHODS

Transparency and Openness

Primary data, demographics, neuropsychological tests, and analysis scripts are publicly available on OSF Framework and Github. Publicly shared data use generic participant labels rather than study IDs. No aspect of the experiment was preregistered.

Participants

Participants were recruited from the Tucson, AZ community, through The Arizona Alzheimer's Prevention Registry and department advertisements. Participants were eligible if they were between 50 and 89 years of age. Exclusion screening criteria included low proficiency in English, a history of neurological disorder, a history of moderate/severe head injury, performance below normative cutoffs on neuropsychological tests using a profile-based approach (Bondi et al., 2014), and safety concerns during their MRI screening (e.g., magnetic bodily implants, claustrophobia, etc.).

Sixty-seven participants were initially enrolled. Based on an a priori exclusion criterion requiring recall of at least six of the eight scenes, ten participants were excluded. Our final sample included 57 older adults (M = 68.77, SD = 6.11, Range = 60-81, 35 Female). Participants self-identified as White (n = 53), Black/African American (n = 2), Asian (n = 1), and Hispanic/Latino (n = 1).

Participants provided written informed consent in accordance with the Institutional Review Board of the University of Arizona, Protocol ID #1607703006. Participants were compensated $10 per hour, with an additional $5 for travel expenses if testing occurred in-person. Participants received an additional $40 for completing the MRI scan. If any sessions were completed virtually, compensation was provided in the form of a Visa, Amazon, or Walmart gift card.

Procedure

Participants completed a total of three to four study visits as part of the broader study. The present analyses focused on data collected during two sessions that were one-week apart. Some study visits were conducted remotely via Zoom for Health due to the COVID-19 pandemic.

Neuropsychological Tests. Participants completed a battery of neuropsychological tasks to test for cognitive impairment (Table 1). These tasks included the Wechsler Adult Intelligence Scale, Fourth Edition (WAIS-IV; Wechsler, 2009) Vocabulary and Similarities to assess verbal comprehension; California Verbal Learning Test, Second Edition (CVLT-II; Delis et al., 2000) to assess episodic verbal learning and memory; Rey Complex Figure copy and recall conditions (Rey, 1941) to assess visuospatial skills and memory; Trail Making Test Part A and B (Reitan & Wolfson, 1985) to assess speed of information processing and executive functions; WAIS-IV Block Design (Wechsler, 2009) to assess visuospatial skills; and Animal Fluency (Benton, 1968) and Boston Naming Task (Goodglass et al., 2001) to assess language.

Encoding. Participants viewed eight 40-second panoramic videos depicting a variety of real-world environments (e.g., a hotel lobby, stadium, distillery, and coastal setting) along with unique titles (Fig. 1A) and were instructed to remember the scenes for a later memory test. Each video was created from a static 360° panoramic image obtained from Google Street View and other publicly posted advertisements. For experimental purposes, each clip was standardized such that the viewpoint panned continuously around the panoramic scene, allowing participants to see different portions of the environment while the people, objects, and other elements within the scene remained static. Thus, changes over the course of each video reflected movement of the viewpoint through the panorama rather than changes occurring within the depicted environment.

Immediate Recall. Participants were cued with the title of each scene, presented in random order, and instructed to picture the scene in their mind and tell the experimenter everything they remember about the scene (Fig. 1B). Participants were provided a maximum time of four minutes and probed to continue if they stopped recalling before two minutes had elapsed.

Delayed Recall. One week later, participants completed the same recall procedure as Immediate Recall. Scenes were presented in random order, and participants were cued with the same title and instructed to picture the scene in their mind and tell the experimenter everything they remember about the scene. Participants were given a maximum time of four minutes and probed to continue if they stopped recalling before two minutes had elapsed.

MRI Acquisition. Participants completed an MRI screening form prior to scanning to confirm safety. Imaging was conducted at the Brain & Body Imaging Center at the University of Arizona on a Siemens 3T Skyra (syngo MR E11) with a 32-channel head coil. Participants were given MRI-compatible foam earbuds and additional pads were placed by their ears to protect hearing and further reduce motion. Participants were also given a foam cushion for under their knees, blankets, and MRI safe glasses as needed. Participants were instructed to hold onto a squeeze ball and squeeze it if they needed to be removed from the MRI immediately.

A T1-weighted 3D MPRAGE anatomical image was acquired in the sagittal plane (TR = 2300 ms, TE = 2.95 ms, TI = 900 ms, flip angle = 9°, matrix = 256 × 240, 176 slices, voxel size = 1.0 × 1.055 × 1.055 mm, slice thickness = 1 mm, pixel bandwidth = 240 Hz/pixel, phase FOV = 93.75%). Resting-state and task-based functional runs were also acquired as part of the broader study protocol but were not analyzed here.

Recall Measures

Our goal was to characterize both the overall episodic composition of recall and the specific scene content that remained represented across a one-week delay. We used detail-based scoring (Levine et al., 2002; Fig 1C) to characterize the composition of recalled content and used text-based similarity measures (Cer et al., 2018; Fig. 1D) to characterize the consistency of scene recall relative to the original scene, to what other participants recalled, and to an individual's own recall across sessions.

For the text-based similarity measures, we used the Universal Sentence Encoder (USE; Cer et al., 2018) to quantify the semantic content of participants' scene descriptions. Each scene recall was converted into 512-dimensional embedding vectors, and cosine similarity was used to quantify the similarity between two pairs of descriptions. We used these embeddings to derive three complementary measures of the consistency of remembered scene content: Description Similarity, Group Similarity, and Group Consistency.

Internal Detail Proportion. We used the Autobiographical Interview scoring protocol (Levine et al., 2002) to characterize the internal and external details produced during recall. For each trial, the transcript was segmented into individual details, then tagged by detail type. Internal Details included event, time, place, perceptual, and thought/emotion details whereas External Details included semantic information, details external to the target scene, and metacognitive statements. We additionally included a False Memory detail type for details describing information that was clearly not a part of the scene. All raters completed a standardized training procedure using a training set before scoring study data and achieved excellent reliability for internal and external details (Cronbach's alpha > .90). For trials scored by two raters, the number of details within each detail type was averaged across raters, and these averaged values were used in subsequent analyses. Although all detail types were coded (Table 2), the present analyses focused

on Internal Detail Proportion, calculated by dividing the number of Internal Details by the total number of details produced (Fig. 1C). This measure characterized the episodic recall composition of recall, with higher Internal Detail proportion values indicating a greater proportion of the participant's recall consisted of Internal Details.

Description Similarity. To quantify the consistency of recalled scene content with the original scene, we compared each recall description with reference descriptions generated by OpenAI GPT-5.6 Sol and Anthropic Claude Opus 5 (Fig. 1E). Similarity was calculated separately for each reference description and then averaged across AI-models. Higher Description Similarity reflected greater semantic correspondence between a participant's recall and the AI-generated reference descriptions of the original scenes.

Consensus Similarity. To characterize the consistency of recalled scene content across individuals, we calculated the pairwise similarity between each participant's recall description and the descriptions produced by the remaining participants for that same scene and same recall session (Fig. 1F). Thus, immediate recall was compared with other participants' immediate recall, and delayed recall was compared with other participants' delayed recall. Pairwise similarity values were then averaged to obtain a Consensus Similarity score, where higher values indicated recall that was more similar to that of the larger group.

Self-Consistency. Finally, we examined the consistently of an individual's recalled scene content across the one-week delay by calculating the similarity between their Session 1 and Session 2 descriptions of the same scene (Fig. 1G). Higher values indicated greater consistency in recall across sessions.

Volumetric Measures

Regions of Interest. MTL volumes were derived using Automatic Segmentation of Hippocampal Subfields (ASHS; Yushkevich et al., 2015), a multi-atlas segmentation approach that registers manually labeled atlas images to each participant's anatomical image and combines the transformed labels via joint label fusion to derive participant-specific segmentations. We used the ASHS-PMC-T1 atlas (Xie et al., 2019), developed for 3T T1-weighted imaging, which includes explicit modeling of the dura to improve segmentation of medial temporal lobe cortical regions. This atlas yields labels for anterior and posterior hippocampus, entorhinal cortex, Brodmann areas 35 and 36, and parahippocampal cortex. Hippocampal volume was computed as the sum of anterior and posterior hippocampus, and perirhinal volume as the sum of BA35 and BA36. Total intracranial volume was estimated by ASHS. Volumes were taken from the "corr_nogray" output in native T1-weighted space.

Quality Control. All ASHS segmentations were visually inspected in ITK-SNAP by two independent raters (A.I.D. and A.N.G.), with each participant's T1-weighted image loaded as the main image and the multi-label segmentation as an overlay. Each hemisphere was rated as Pass, Check, or Fail. Fail indicated gross registration failure or mislabeling severe enough to render regional volumes uninterpretable, and Check indicated minor boundary imprecision judged unlikely to substantially affect volumes. No segmentations were rated as Fail, and no manual editing was necessary.

MTL Composite. To derive a composite measure of medial temporal lobe (MTL) volume, volumes for the hippocampus, entorhinal cortex, perirhinal cortex, and parahippocampal cortex were summed across hemispheres and standardized across participants. Following similar approaches used in prior work (Bailey et al., 2013; Pettigrew et al., 2017), each regional volume was standardized across participants and separately adjusted for intracranial volume (ICV) by

regressing the standardized volume on ICV and retaining the residual. The ICV-adjusted volumes were averaged across the four regions to produce a single MTL composite, with higher values reflecting relatively greater MTL volume after accounting for ICV.

Statistical Analyses

Statistical analyses were conducted in R (version 4.5.2). Linear mixed-effects models were fit using the lme4 package (Bates et al., 2015), and p-values for fixed effects were obtained with lmerTest (Kuznetsova et al., 2017) using Satterthwaite approximations for degrees of freedom.

Changes in recall across sessions. We first examined how each measure of scene recall changed across the one-week delay. Separate linear mixed-effects models were fit for Internal Detail Proportion, Description Similarity, and Group Similarity. Each model included session as a fixed effect while controlling for age (mean-centered), with random intercepts for participant and scene:

$$Dependent\ Variable \sim Session + Age + (1|Participant) + (1|Scene)$$

Session 1 was coded as the reference condition, such that the effect of session reflected the change from immediate to delayed recall. Self-Consistency was not included in these analyses because it was calculated from the similarity between Session 1 and Session 2 recall and therefore did not vary by session.

MTL composite and scene memory. Our primary analyses examined whether individual differences in MTL volume were related to scene memory and whether these relationships changed across the one-week delay. Separate linear mixed-effects models were fit for Internal Detail Proportion, Description Similarity, and Group Similarity. Models included MTL composite volume, session, and their interaction while controlling for age, with random intercepts for participant and scene:

$$Dependent\ Variable \sim MTL\ Volume * Session + Age + (1|Participant) + (1|Scene)$$

Session 1 was again coded as the reference condition. Thus, the main effect of MTL volume reflected its relationship with immediate recall, whereas the MTL × Session interaction tested whether this relationship differed at delayed recall. When an interaction was observed, we estimated the relationship between MTL volume and the dependent measure separately at each session.

Given that Self-Consistency reflected similarity across the two recall sessions, it was examined in a separate model without a session term:

$$Self\ Consistency \sim MTL\ Volume + Age + (1|Participant) + (1|Scene)$$

Analyses involving the MTL composite were hypothesis-driven and were not corrected for multiple comparisons.

Exploratory MTL subregion analyses. We next explored whether the relationships observed for the MTL composite were evident in specific MTL regions. Separate models were fit for hippocampal, entorhinal cortex, perirhinal cortex, and parahippocampal cortex volumes using the same model structure as the primary analyses:

$$Dependent\ Variable \sim Subregion\ Volume * Session + Age + (1|Participant) + (1|Scene)$$

For each dependent measure, p-values for the Subregion Volume × Session interaction were corrected across the four MTL subregions using the Benjamini-Hochberg false discovery rate procedure (Benjamini & Hochberg, 1995). For Self-Consistency, separate models included subregional volume and age without a session term, and FDR correction was applied to the main effect of regional volume across the four subregions. FDR-adjusted p-values below .05 were considered significant ($q < .05$).

RESULTS

Scene recall became longer and less similar across the one-week delay

We first examined how recall changed across the one-week delay. To do so, we fit separate linear mixed-effects models for each session-varying recall measure, examining the effect of session while controlling for age (mean-centered), with random intercepts for participant and scene. Overall, features of scene recall showed distinct patterns of change across the one-week delay.

Internal Detail Proportion did not significantly differ between immediate and delayed recall (b = -0.014, SE = 0.008, t(838.15) = -1.72, 95% CI [-0.030, 0.002], p = .086; Fig. 2). Description Similarity, however, decreased across sessions, indicating that delayed recall was less similar to AI-generated descriptions of the original scenes (b = -0.011, SE = 0.004, t(838.16) = -2.43, 95% CI [-0.019, -0.002], p = .015; Fig. 2). Consensus Similarity also decreased after the one-week delay (b = -0.024, SE = 0.003, t(838.46) = -7.20, 95% CI [-0.031, -0.018], p < .001; Fig. 2), suggesting that participants' delayed descriptions were less similar to those produced by other participants relative to their similarity at immediate recall. In contrast, participants produced more words after the one-week delay (b = 50.96, SE = 7.63, t(837.06) = 6.68, 95% CI [36.00, 65.93], p < .001).

Importantly, age was not significantly associated with Internal Detail Proportion (b = -0.005, SE = 0.003, t(55.03) = -1.56, p = .124) or Description Similarity (b = -0.002, SE = 0.001, t(55.02) = -1.73, p = .089), though older age was associated with lower Consensus Similarity (b = -0.0014, SE = 0.0006, t(55.15) = -2.27, p = .027).

MTL-memory relationships changed after the one-week delay

We next examined whether individual differences in MTL volume were related to scene memory and whether these relationships changed across the one-week delay. Separate linear

mixed-effects model included MTL volume, session, and their interaction while controlling for age (mean-centered), with random intercepts for participant and scene for each of our dependent variables including Description Similarity, Group Similarity, Internal Detail Proportion, and Self-Consistency. Overall, we found that the relationship between MTL volume and memory differed across memory measures and varied over a one-week delay.

Relationship between MTL volume and Internal Detail Proportion weakened across the delay. There was no significant main effect of MTL volume (b = 0.035, SE = 0.021, t(59.21) = 1.67, p = .100) or Session (b = -0.013, SE = 0.008, t(837.14) = -1.57, p = .117). However, there was a significant interaction between MTL volume and Internal Detail Proportion (b = -0.026, p = .003; Fig. 3), indicating that the relationship between MTL volume and Internal Detail Proportion weakened across the delay. However, MTL volume was not significantly associated with Internal Detail Proportion at immediate recall (b = 0.035, 95% CI [-0.007, 0.078], p = .100) or at delayed recall (b = 0.009, 95% CI [-0.033, 0.052], p = .657).

Description Similarity and Consensus Similarity were associated with MTL volume after the one-week delay. For Description Similarity, there was no significant main effect of MTL volume (b = 0.012, SE = 0.009, t(63.27) = 1.34, p = .184). There was, however, a significant main effect of Session (b = -0.011, SE = 0.004, t(837.15) = -2.57, p = .010), reflecting lower Description Similarity after the one-week delay. There was also a significant interaction between MTL volume and Description Similarity (b = 0.011, SE = 0.005, t(837.27) = 2.37, p = .018; Fig. 3). MTL volume was not significantly associated with Description Similarity at immediate recall (b = 0.012, 95% CI [-0.006, 0.029], p = .184), but greater MTL volume was associated with higher Description Similarity after the one-week delay (b = 0.023, 95% CI [0.006, 0.041], p = .011), suggesting that individuals with better retention of the original scenes tended to have greater MTL volume.

Consensus Similarity showed a comparable pattern. There was no significant main effect of MTL volume (b = 0.006, SE = 0.005, t(76.60) = 1.36, p = .179), whereas the main effect of Session was significant (b = -0.024, SE = 0.003, t(837.50) = -7.32, p < .001), reflecting lower Consensus Similarity after one week. The MTL × Session interaction, while not significant, was exactly at the alpha significance threshold (b = 0.007, SE = 0.004, t(837.77) = 1.96, p = .050; Fig. 3). Aligning with this result, MTL volume was not significantly associated with Consensus similarity at immediate recall (b = 0.006, 95% CI [-0.003, 0.015], p = .179), whereas greater MTL volume was associated with higher Consensus Similarity after one week (b = 0.013, 95% CI [0.004, 0.022], p = .005). This indicates that individuals who tended to retain a normative description of the scenes tended to have larger MTL volume.

Greater MTL volume was associated with more consistent recall across sessions. We next asked whether MTL volume was related to the consistency of each participant's recall across the two sessions. Given that Self-Consistency was calculated by comparing immediate and delayed recall, this model did not include session as a predictor. We found greater MTL volume was associated with higher Self-Consistency (b = 0.024, SE = 0.008, t(55.38) = 2.96, p = .005; Fig. 3), such that participants with larger MTL volumes produced more similar descriptions of the same scene across the one-week interval.

MTL volume was unrelated to words recalled. Although word count was not a primary measure of interest, we also examined the number of words produced during recall to assess whether the observed relationships might reflect differences in verbal output rather than remembered scene content. There was no significant main effect of MTL volume on Word Count (b = 9.37, SE = 30.40, t(56.18) = 0.31, p = .759). The main effect of Session was significant (b = 51.47, SE = 7.64, t(836.05) = 6.74, p < .001), reflecting greater verbal output after the one-week

delay. The MTL Volume × Session interaction was not significant (b = -10.85, SE = 8.19, t(836.11) = -1.32, p = .186; Fig. 3).

Lastly, age was not significantly associated with any measure in these models (Internal Detail Proportion, p = .352; Description Similarity, p = .477; Consensus Similarity, p = .266; Self-Consistency, p = .529; Word Count, p = .501).

Distinct MTL subregions tracked different features of scene memory

To determine whether the relationships observed for the MTL composite were associated with specific subregions, we next examined the hippocampus, entorhinal cortex, perirhinal cortex, and parahippocampal cortex separately. For each dependent variable, we fit linear mixed-effects models including subregion volume, session, and their interaction while controlling for age, with random intercepts for participant and scene. Given that these analyses were exploratory, p-values for the Subregion Volume × Session interaction were FDR-corrected ($q < .05$) across the four subregions within each dependent measure. For Self-Consistency, which was calculated across sessions, FDR correction was applied to the main effect of subregion volume. Overall, the exploratory analyses showed that the MTL composite relationships were not uniform across regions.

Relationship between entorhinal volume and Internal Detail Proportion changed across sessions. There was no significant overall main effect of entorhinal volume (b = -0.008, SE = 0.018, t(54.10) = -0.43, p = .673) or Session (b = -0.014, SE = 0.008, t(837) = -1.73, p = .083) for Internal Detail Proportion. However, the Entorhinal Volume × Session interaction was significant (b = -0.030, SE = 0.008, t(837) = -3.69, p < .001, q = .001), reflecting a change in the relationship across the one-week delay. However, entorhinal volume was not significantly associated with Internal Detail Proportion during either immediate recall (b = 0.007, 95% CI [-0.030, 0.044], p =

.701) or delayed recall (b = -0.023, 95% CI [-0.060, 0.015], p = .229). Thus, the interaction effect appears to reflect a change in association across sessions rather than a significant relationship at either session alone.

This pattern did not extend to the similarity-based measures. Entorhinal volume was not significantly associated with Description Similarity (b = 0.001, SE = 0.008, t(54.10) = 0.11, p = .917, q = .917) or Consensus Similarity (b = 0.002, SE = 0.004, t(54.30) = 0.63, p = .534, q = .534), and did not show significant Session interactions for Description Similarity (b = -0.001, SE = 0.004, t(837) = -0.13, p = .894, q = .894) or Consensus Similarity (b = 0.002, SE = 0.003, t(838) = 0.59, p = .555, q = .740). Description Similarity and Consensus Similarity nevertheless decreased across sessions (ps = .015 and < .001, respectively). Entorhinal volume was also not significantly associated with Self-Consistency (b = 0.007, SE = 0.008, t(54.8) = 0.90, p = .370, q = .376).

Perirhinal volume was related to similarity-based measures after the one-week delay. Perirhinal cortex showed the most consistent pattern across the similarity-based measures. For Description Similarity, the overall main effect of perirhinal volume did not survive FDR correction (b = 0.017, SE = 0.007, t(54.00) = 2.50, p = .015, q = .062), whereas the main effect of Session was significant (b = -0.011, SE = 0.004, t(837) = -2.44, p = .015). The relationship with perirhinal volume differed across sessions (b = 0.012, SE = 0.004, t(837) = 2.77, p = .006, q = .023). Perirhinal volume was not significantly associated with Description Similarity during immediate recall (b = 0.011, 95% CI [-0.003, 0.026], p = .123), whereas greater volume was associated with higher Description Similarity after the one-week delay (b = 0.023, 95% CI [0.009, 0.037], p = .002).

Consensus Similarity showed a similar pattern. The overall main effect of perirhinal volume did not survive FDR correction (b = 0.007, SE = 0.003, t(54.10) = 2.12, p = .039, q = .136),

whereas the main effect of Session was significant (b = -0.024, SE = 0.003, t(838) = -7.23, p < .001). The Perirhinal Volume × Session interaction was significant (b = 0.008, SE = 0.003, t(837) = 2.58, p = .010, q = .040). Perirhinal volume was not significantly associated with Consensus Similarity during immediate recall (b = 0.003, 95% CI [-0.004, 0.011], p = .409), but greater perirhinal volume was associated with higher Consensus Similarity at delayed recall (b = 0.012, 95% CI [0.004, 0.019], p = .004).

Perirhinal volume was also associated with Self-Consistency (b = 0.024, p = .0008, q = .003), such that participants with greater volume produced more similar descriptions of the same scene across sessions. In contrast, perirhinal volume did not show significant interaction with session for Internal Detail Proportion (b = -0.003, SE = 0.008, t(837) = -0.32, p = .753, q = .753).

Hippocampal effects did not survive correction. None of the Hippocampal Volume × Session interactions survived FDR correction. The interactions for Description Similarity (b = 0.011, SE = 0.005, t(837) = 2.16, p = .031, q = .062) and Consensus Similarity (b = 0.008, SE = 0.004, t(838) = 2.05, p = .040, q = .081) were significant prior to correction but did not remain significant after accounting for multiple comparisons. The interaction for Internal Detail Proportion was also not significant (b = -0.009, SE = 0.009, t(837) = -1.00, p = .317, q = .423). Main effects of hippocampal volume likewise did not survive FDR correction for Internal Detail Proportion (b = 0.039, SE = 0.021, t(54.1) = 1.89, p = .065, q = .258), Description Similarity (b = 0.012, SE = 0.009, t(54.1) = 1.31, p = .196, q = .392), or Consensus Similarity (b = 0.008, SE = 0.004, t(54.2) = 1.86, p = .068, q = .136). Hippocampal volume was also not significantly associated with Self-Consistency (b = 0.012, SE = 0.009, t(54.4) = 1.32, p = .192, q = .376).

Parahippocampal volume was not significantly related to the scene-memory measures after correction. There were no significant main effects of parahippocampal volume for Internal

Detail Proportion (b = 0.014, SE = 0.019, t(54.1) = 0.76, p = .453, q = .673), Description Similarity (b = 0.006, SE = 0.008, t(54.1) = 0.82, p = .415, q = .554), or Consensus Similarity (b = 0.003, SE = 0.004, t(54.2) = 0.72, p = .478, q = .534). The Parahippocampal Volume × Session interaction for Internal Detail Proportion was significant prior to correction but did not survive FDR correction (b = -0.018, SE = 0.008, t(837) = -2.22, p = .026, q = .053). The Volume × Session interactions were also not significant for Description Similarity (b = 0.005, SE = 0.004, t(837) = 1.14, p = .254, q = .339) or Consensus Similarity (b = -0.001, SE = 0.003, t(838) = -0.29, p = .776, q = .776). Parahippocampal volume was also not significantly associated with Self-Consistency (b = 0.007, SE = 0.008, t(54.5) = 0.89, p = .376, q = .376).

DISCUSSION

Scenes are thought to provide the spatial and contextual framework in which everyday events occur, but in naturalistic experiences, they are embedded within other information that unfolds across the event. Although the MTL has been implicated in reconstructing complex events, less is known about how the MTL relates to the retention of detailed scene information over time. To address this question, we isolated scene memory using 360-degree panoramic videos of real-world environments that participants recalled shortly after viewing and again one week later. Consistent with our hypothesis, individual differences in MTL volume were associated with multiple features of scene memory, with this relationship emerging only after a one-week delay. Together, these findings provide evidence that the MTL may play a role in retaining the details of complex scenes over time, ensuring that the stage upon which event memories unfold is set for an elaborative reconstruction of the past.

Across the one-week delay, scene recall changed in several ways. Participants produced more words, and their descriptions became less similar to descriptions of the original scenes, as well as to those produced by other participants. Interestingly, the proportion of internal or episodic details did not significantly change across the delay. This dissociation suggests that the overall episodic composition of recall remained relatively stable even as the specific scene information being retained changed over the week. Prior work similarly shows that naturalistic memories are selectively retained, with central information remaining relatively stable while peripheral details are more likely to be lost over time (Sekeres et al., 2016). Our findings extend this work to scenes, where there is no unfolding event structure that defines which information is central or peripheral. Although individuals may have imposed their own organizational strategies during intentional encoding, such self-generated structure would likely vary across individuals, leaving greater room

for individuals to focus on different features of the same environment and, over time, to retain different details of the same scene.

The selective changes in retained scene content were accompanied by changes in its relationship with MTL. At immediate recall, MTL volume was not reliably associated with how closely recall corresponded to the original scenes or to what others recalled. In contrast, greater MTL volume was associated with both features of scene memory after one week and with greater self-consistency across sessions. Thus, individual differences in MTL became particularly informative about the scene content that remained represented after the delay, whether retention was assessed relative to the original scene, to what other participants recalled, or to an individual's own earlier memory. This pattern is consistent with prior work indicating that MTL volume is associated with memory for short stories after one week but not at shorter delays (Saloner et al., 2018). The present findings extend this work to naturalistic scenes that were viewed once and later freely reconstructed from memory, showing that this memory retention relationship with MTL structure is also evident for the scene component of an experience in the absence of an unfolding narrative.

Importantly, the emergence of MTL-memory relationships after a delay may reflect changes in the memory representation rather than changes in whether the MTL supports memory. Immediately after encoding, many features of a scene may remain accessible through non-MTL specific cognitive processes (e.g., mental rehearsal), and individual differences in structural volume of the MTL may therefore provide relatively little leverage for distinguishing among individuals. Across the week, however, as fewer features remain available and memories becomes more selective, stable individual differences in MTL volume may become increasingly informative about which scene details are retained and available for later reconstruction. This interpretation is

broadly compatible with memory models proposing that memory representations are reorganized or transformed over time (Winocur & Moscovitch, 2011; Squire et al., 2015), although the present findings cannot distinguish among specific memory accounts.

Although the MTL volumetric composite provided the primary evidence for these delayed relationships, prior models suggest that the MTL's constituent regions contribute differently to the information that makes up a remembered scene. According to these models, the hippocampus, perirhinal cortex, entorhinal cortex, and parahippocampal cortex participate in partially distinct but interacting memory pathways, with different regions contributing more strongly to item information, contextual information, and the relationships among them (Davachi, 2006; Diana et al., 2007; Ranganath & Ritchey, 2012). Because scene memory draws on each of these features, the composite MTL finding may reflect contributions distributed across this broader system rather than a process localized to a single neural region with in the MTL. Exploratory regional analyses provided some support for this possibility. Hippocampal effects for the similarity measures were in a similar direction as the composite findings but did not survive correction for multiple comparisons. Entorhinal cortex showed a session-dependent relationship with episodic detail composition, whereas parahippocampal cortex showed no reliable relationships with the scene-memory measures.

Within this broader regional pattern, perirhinal cortex showed the clearest relationship with scene memory, with greater volume associated with retention of scene content relative to the original scenes, what others recalled, and an individual's own earlier memory. One possible account of this pattern follows from the broader organization of MTL memory systems, in which perirhinal cortex is especially important for representing specific objects or items, while other MTL regions contribute more strongly to contextual and relational information (Cohen and

Eichenbaum, 1993). Although scenes may have a context-setting quality to them, that context may be inherently supported by the objects or items that are present. Our similarity measures therefore may have been particularly sensitive to details about the objects or items retained over the delay, and this emphasis may partly explain the stronger relationship with perirhinal cortex (Bussey et al., 2002; Bussey et al., 2005; Clarke & Tyler, 2014; Frick et al., 2023). These analyses were exploratory, however, and the prominence of perirhinal cortex should be interpreted as one possible clue to the type of scene information contributing to the broader MTL effect rather than as evidence that scene memory depends primarily on this region.

The older adult sample provides an important context for these findings, but the present study cannot determine whether the observed relationships are specific to cognitive and brain aging. Perirhinal cortex and other MTL regions are vulnerable to age-related change and are affected early in the course of Alzheimer's disease pathology (Braak & Braak, 1991, 1995; Berron et al., 2020). This makes individual differences in MTL structure particularly relevant in older adults. At the same time, our participants were cognitively healthy, age itself did not account for our brain-behavior relationships, and we did not include a younger adult comparison group. We therefore cannot determine whether the same relationships between MTL structure and scene memory would also be observed earlier in adulthood or whether they become more pronounced with age-related neural change. Future work comparing age groups and incorporating markers of Alzheimer's disease pathology could help distinguish general features of scene memory from relationships that may become important in aging.

The use of an immediate recall also may have influenced the results of this study. One process that may reinforce individualized scene memory is retrieval itself, as retrieval can strengthen recovered information while also modifying what is subsequently remembered

(Roediger & Karpicke, 2006; Bridge & Paller, 2012). Given that participants recalled each scene shortly after encoding, initial retrieval may have sharpened which details remained available one week later. The relatively high within-individual consistency across sessions aligns with this possibility. That is, although participants became less similar to one another, they tended to reconstruct the same scene in a largely reliable manner. However, without a condition where there is no initial, immediate retrieval, we cannot separate effects of prior retrieval from other changes across the retention interval. Future studies that manipulate whether an initial recall occurs could test whether early retrieval stabilizes an individualized representation of the scene and, in turn, shapes which scene details are retained at a later recall.

Several additional limitations point to important directions for future work. First, volumetric measures characterize individual differences in MTL structure but cannot reveal how these regions are engaged during memory processing. Functional imaging could complement these findings by examining MTL activity while encoding and retrieving complex, naturalistic scenes over time. Second, although our similarity-based approach provides a scalable way to characterize naturalistic recall without extensive manual scoring, correspondence to the original scene was in part based on AI-generated reference descriptions. Although these descriptions were reviewed by experimenters, they may not fully capture the information available in a complex visual scene. Notably, we supplemented this measure with comparisons to other participants' recall and to each participant's own recall across sessions, providing complementary measures of consistency with which scene content was retained. Finally, the central strength of the present design is also an important limitation. That is, we intentionally removed the scene from the broader event in which scenes are normally experienced. Isolating scenes allowed us to identify a relationship between MTL structure and scene memory without the confounding influence of character building,

actions, dialogue, and narrative structure, but everyday scenes typically form part of richer unfolding experiences. Future work should determine how memory for the scene interacts with memory for the people, actions, and events that occur within it.

In summary, these findings demonstrate that the MTL is related to the retention of detailed scene information and that this relationship changes as scene memories become more selective over time. Across one week, scene memories became less similar to the original scenes and to what other people remembered, even though the overall proportion of episodic content remained relatively stable. Individual differences in MTL volume became particularly informative about which details remained represented and how consistently events were reconstructed after the delay. Our findings provide a foundation for understanding how the places in which experiences occur are selectively retained and reconstructed as part of the complex events we later remember.

FIGURES AND TABLES

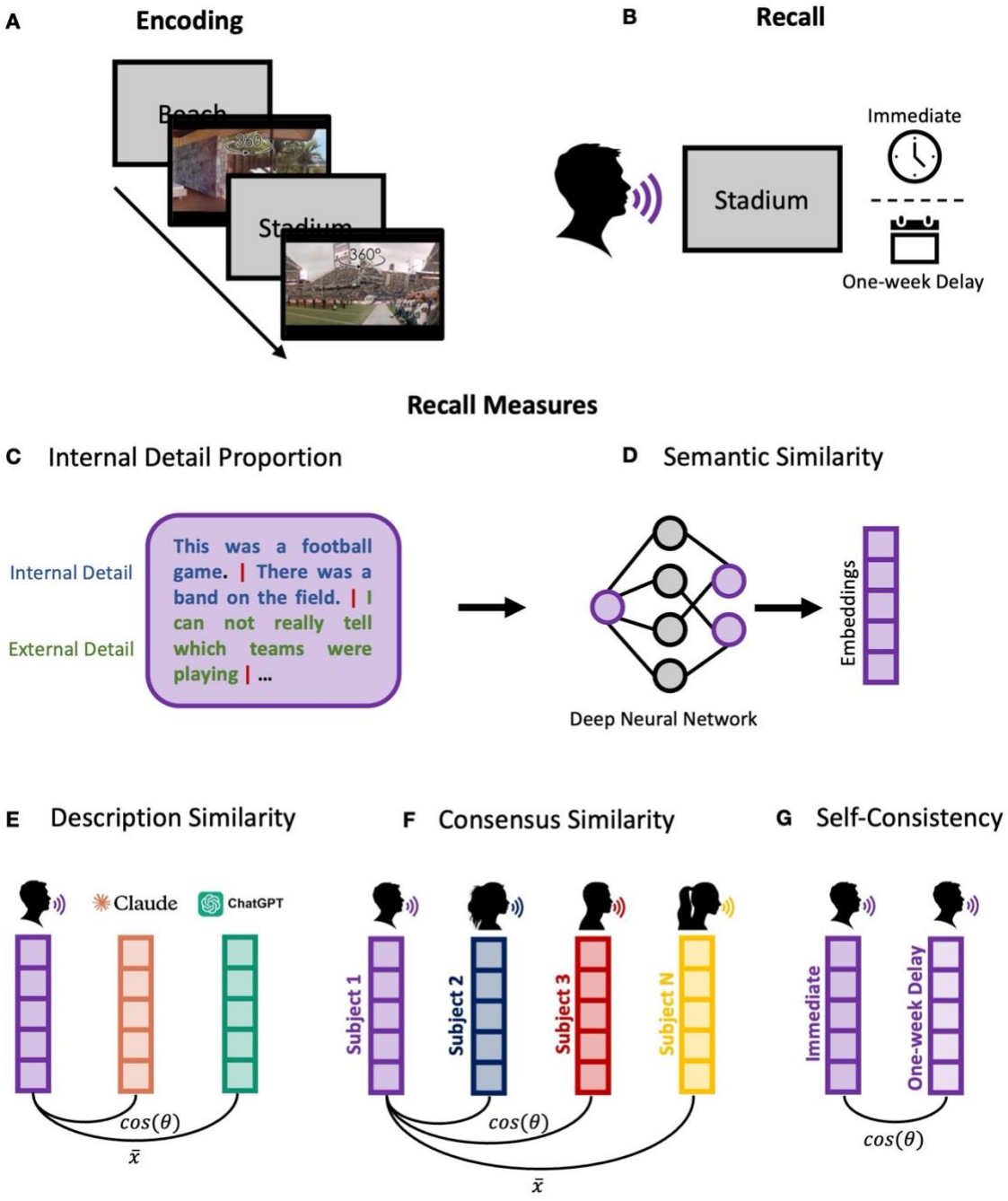


Figure 1. Experimental Methods and Recall Measures. ***(A)*** *During encoding, participants viewed eight 40-second panoramic videos of real-world environments, each preceded by a title. Videos were generated from static 360° panoramic images, with the viewpoint panning continuously across the scene.* ***(B)*** *At recall, participants were cued with the scene title and asked to describe everything they could remember about the environment, shortly after encoding and again after a one-week delay.* ***(C)*** *Each recall was transcribed, segmented into individual details, and scored as internal or external following the Autobiographical Interview protocol (Levine et al., 2002). Internal Detail Proportion was calculated as the number of internal details divided by the total number of details and characterized the episodic composition of recall.* ***(D)*** *Each recall was converted into a 512-dimensional embedding using the Universal Sentence Encoder (Cer et al., 2018), and cosine similarity was used to derive semantic similarity measures characterizing the consistency of remembered scene content.* ***(E)*** *Description Similarity was calculated as the cosine*

similarity between a participant's recall and reference descriptions of the original scene generated by two large language models, averaged across models. ***(F)*** *Consensus Similarity was calculated as the average pairwise cosine similarity between a participant's recall and the recall of all other participants for the same scene and session.* ***(G)*** *Self-Consistency was calculated as the cosine similarity between a participant's recalled scene across the delay.*

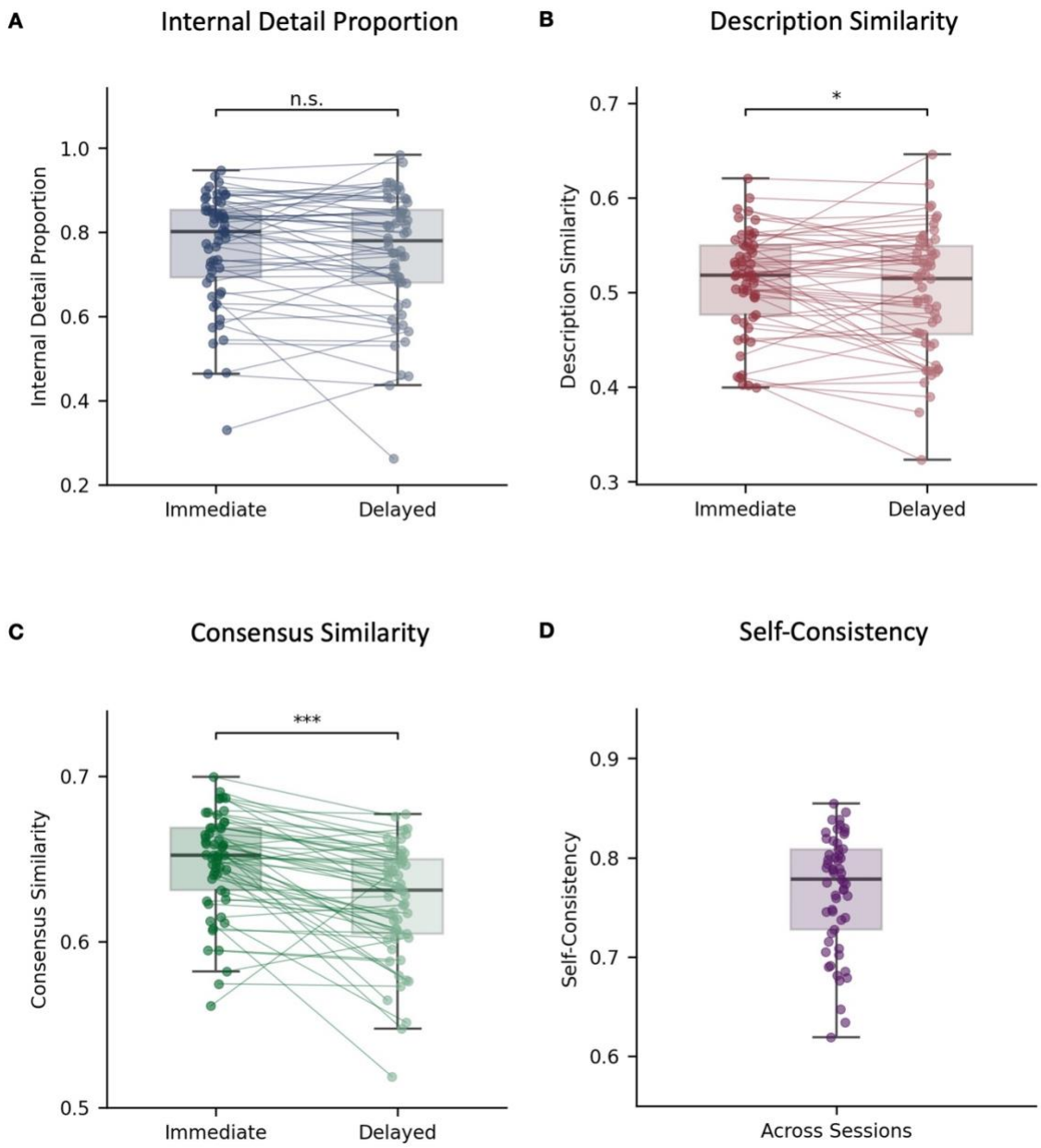


Figure 2. Scene Recall Measures at Immediate and Delayed Testing. *Each panel shows a recall measure at Immediate (Session 1) and Delayed (Session 2) testing.* ***(A)*** *Internal Detail Proportion, the number of internal details divided by the total number of details produced, reflecting the episodic composition of recall* ***(B)*** *Description Similarity, the semantic similarity between a participant's recall and model-generated reference descriptions of the original scene, reflecting consistency with the original scene.* ***(C)*** *Consensus Similarity, the average semantic similarity between a participant's recall and the recall of all other participants for the same scene. Reflecting consistency with what other participants recalled.* ***(D)*** *Self-Consistency, the semantic similarity between a participant's immediate and delayed recall of the same scene, reflecting an individual's recalled scene across the one-week delay. Because this measure is calculated across sessions, it is shown as a single distribution.* ***Key:*** *Points represent participant means averaged across scenes, and lines connect the same participant across sessions. Boxes show the interquartile range with the median indicated by the horizontal line; whiskers extend to 1.5 times the interquartile range. n.s. = not significant. *p < .05. ***p < .001.*

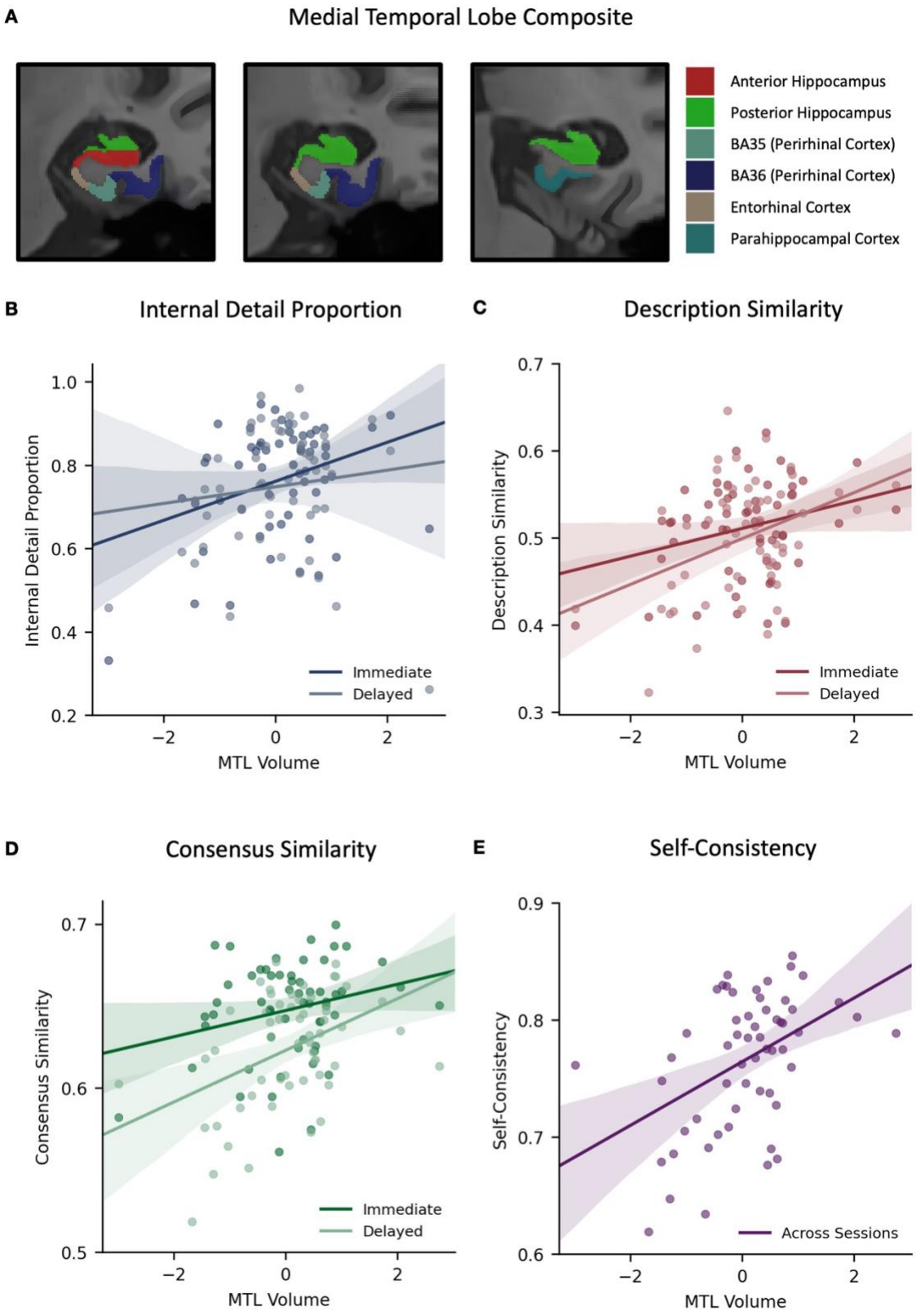


Figure 3. Medial Temporal Lobe Volume and Scene Recall at Immediate and Delayed Testing. ***(A)*** *Representative ASHS segmentation of medial temporal lobe subregions in a single participant, shown on coronal slices of the T1-weighted anatomical*

image. The MTL composite was calculated by summing volumes across hemispheres for the hippocampus, entorhinal cortex, perirhinal cortex, and parahippocampal cortex, standardizing each regional volume, adjusting for intracranial volume, and averaging across regions. ***(B–E)*** *Relationships between MTL composite volume and each recall measure:* ***(B)*** *Internal Detail Proportion,* ***(C)*** *Description Similarity,* ***(D)*** *Consensus Similarity, and* ***(E)*** *Self-Consistency.* ***Key:*** *Points represent participant means averaged across scenes; darker and lighter shading distinguishes Immediate (Session 1) from Delayed (Session 2) recall. Lines show model-estimated associations at each session with 95% confidence bands. Self-Consistency was calculated by comparing immediate and delayed recall and therefore is shown as a single relationship across sessions. Higher MTL composite values reflect relatively greater medial temporal lobe volume after accounting for intracranial volume. BA = Brodmann area.*

Table 1. Neuropsychological Tests. *Values are age-adjusted z-scores. Higher values indicate better performance; Trail Making Test scores were reverse-scored so that higher values reflect faster completion times. WAIS-IV Verbal Comprehension Index was converted from the standard index score (M = 100, SD = 15). CVLT-II = California Verbal Learning Test, Second Edition; WAIS-IV = Wechsler Adult Intelligence Scale, Fourth Edition.*

Neuropsychological Test	*M (SD)*	*Min*	*Max*
CVLT-II Long Delay Free Recall	0.68 (0.99)	-1.5	2.5
CVLT-II Trials 1-5 Total	0.90 (0.93)	-1.2	2.7
Boston Naming Test	0.91 (0.90)	-1.0	2.4
Animal Fluency	0.40 (0.92)	-1.4	2.5
Trail Making Test A	0.06 (0.93)	-2.0	3.0
Trail Making Test B	0.16 (0.96)	-1.8	2.5
WAIS-IV Verbal Comprehension Index	1.63 (0.76)	0.00	3.13

Table 2. Autobiographical Interview Detail Scores at Immediate and Delayed Recall. *Values are participant means averaged across the eight scenes within each session. Recall transcripts were segmented into individual details and classified using the Autobiographical Interview scoring protocol (Levine et al., 2002). Internal details describe the specific scene and include event, place, time, perceptual, and thought/emotion details. External details include semantic information, repetitions, and metacognitive statements. False memories were scored when a participant described information that was clearly not part of the scene and were not counted toward either internal or external totals. Internal Detail Proportion was calculated as the number of internal details divided by the total number of details. A subset of transcripts was scored independently by two raters; for these transcripts, detail counts were averaged across raters, which is why some values are non-integer.*

Autobiographical Interview Details		**Immediate**			**Delayed**		
Category	**Detail Type**	*M (SD)*	*Min*	*Max*	*M (SD)*	*Min*	*Max*
Internal Details	Event	7.29 (2.26)	1.62	11.62	7.82 (2.98)	2.75	15.5
	Place	3.10 (2.86)	0	9.31	3.13 (3.12)	0	11.25
	Time	0.25 (0.30)	0	1.57	0.20 (0.29)	0	1.19
	Perceptual	12.88 (5.92)	1.88	30.25	12.37 (7.37)	2.81	36.75
	Thought/Emotion	1.53 (1.45)	0	6	1.76 (1.56)	0	7.5
External Details	Semantic	0.47 (1.03)	0	9	0.42 (1.11)	0	8
	Repetition	1.68 (1.25)	0	5.75	1.64 (1.23)	0	6.5
	Metacognitive	5.15 (3.61)	0.38	16.75	4.77 (3.18)	0.12	12.88
False Memories	False Memories	0.58 (0.82)	0	3	0.79 (0.94)	0	4.12
Total	Total Internal Details	25.05 (10.38)	7.62	55.5	25.28 (12.37)	7.06	63.88
	Total External Details	7.36 (4.45)	0.75	20.38	6.99 (4.14)	0.25	16.75
	Total Details	32.42 (11.24)	9.62	66.88	32.27 (13.33)	9.31	77
	Internal Detail Proportion	0.76 (0.13)	0.33	0.95	0.75 (0.15)	0.26	0.98

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AUTHOR CONTRIBUTIONS

Angelique I. Delarazan: Conceptualization, data curation, formal analysis, investigation, methodology, project administration, software, supervision, visualization, and writing (original draft, review, and editing). **Alexis N. Garcia**: Conceptualization, data curation, formal analysis, investigation, methodology, and writing (original draft, review, and editing). **Ashley A. Huggins**: Methodology, validity, and writing (review and editing). **Matthew D. Grilli**: Conceptualization, funding acquisition, investigation, methodology, project administration, resources, supervision, and writing (original draft, review, and editing).

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